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DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

MINI PROJECT

Submitted by

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**AI23331 - FUNDAMENTALS OF
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BONAFIDE CERTIFICATE

Certified that this project report titled “**Resume Skill Analyzer and Job Recommendation System**” is the Bonafide work of **SANJEEVE R V (231701048)** who carried out the project work for the course **AI23331 - Fundamentals of Machine Learning** under my supervision.

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MINI PROJECT REPORT
CREDIT CARD FRAUD DETECTION USING MACHINE LEARNING

1. INTRODUCTION

Many job seekers find it difficult to identify suitable job opportunities based on their current skill set. They may also be unaware of the skills required for specific job roles and the gap between their existing skills and industry expectations. This creates a need for an intelligent system that can analyze resumes, extract skills, compare them with job descriptions, and recommend relevant job positions.

The AI-Based Resume Skill Analyzer and Job Recommendation System uses Natural Language Processing and machine learning techniques to process resume content and job posting data. The system extracts important skills from the resume, matches them with required job skills, calculates similarity scores, and provides skill gap analysis to guide users toward career improvement.

This project is useful for:

- Helping job seekers identify suitable job roles
- Reducing time spent manually searching for jobs
- Understanding missing skills for desired job positions
- Providing personalized career guidance using machine learning

2. OBJECTIVES

The objectives of this project are:

- To automatically extract skills from uploaded resumes
- To match extracted skills with job descriptions and required skill lists
- To recommend the most relevant job positions using TF-IDF and cosine similarity
- To identify missing skills for selected or recommended job roles
- To build a simple Streamlit web application for user interaction
- To help users understand what skills they need to improve for better job eligibility

3. METHODOLOGY

The methodology of this project is structured into the following stages:

3.1. Data Acquisition

The system uses a job posting dataset containing job title, company name, location, job description, and required skills. A sample dataset is created in CSV format for testing. This dataset can be replaced with a Kaggle job posting dataset by mapping the required columns to the project format.

The job dataset contains the following fields:

- Job title
- Company
- Location
- Job description
- Required skills

3.2. Resume Text Extraction

The user can upload a resume in PDF, DOCX, or TXT format. The system extracts raw text from the uploaded document. PDF resumes are processed using pdfminer.six, DOCX resumes are processed using docx2txt, and TXT resumes are read directly. The extracted text becomes the input for skill extraction and job matching.

3.3. Skill Extraction

A predefined skill keyword list is used to identify technical and professional skills in the resume. The resume text is normalized by converting it to lowercase, removing unnecessary symbols, and standardizing whitespace. Each skill keyword is then matched against the normalized resume text.

Examples of skills used in the system include Python, SQL, Machine Learning, Deep Learning, NLP, Pandas, NumPy, Scikit-learn, TensorFlow, Flask, Streamlit, React, Docker, AWS, Power BI, and Tableau.

3.4. TF-IDF Vectorization

TF-IDF stands for Term Frequency-Inverse Document Frequency. It converts resume text and job descriptions into numerical vectors based on word importance. Words that occur frequently in one document but not across all documents receive higher importance. This helps the system compare resume content with job descriptions mathematically.

3.5. Cosine Similarity Matching

Cosine similarity is used to calculate the similarity between the resume vector and each job description vector.

The similarity score ranges from 0 to 1, where a higher value indicates a better match. The scores are converted into percentages and sorted to recommend the most relevant jobs to the user.

3.6. Skill Gap Analysis

Skill gap analysis is performed using set difference. The system compares the user's extracted resume skills with the required skills of a selected job role.

Skills present in the job requirement but missing from the user's resume are displayed as missing skills. This helps the user understand what to learn next.

3.7. Web Application Development

A Streamlit web application is developed to provide a simple and interactive interface. The user can upload a resume, paste resume text, choose the number of recommendations, select a desired job role, and view extracted skills, recommended jobs, matched skills, and missing skills.

4. CODE

4.1. Import Libraries

All required libraries for data processing, NLP-style text processing, machine learning, and web application development are imported:

```
from pathlib import Path
import pandas as pd
import streamlit as st
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
from pdfminer.high_level import extract_text
import docx2txt
```

4.2. Load Dataset and Skill Datasets

The job posting dataset and skill keyword list are loaded from the data folder:

```
jobs_df = pd.read_csv("data/jobs.csv")
with open("data/skills.txt", "r", encoding="utf-8") as file:
    skill_keywords = [line.strip() for line in file if line.strip()]
```


4.3. Text Normalization

Text normalization converts the resume and job descriptions into a clean format for matching:

```
import re
def normalize_text(text):
    text = text.lower()
    text = re.sub(r"[^a-z0-9+#\s-]", " ", text)
    text = re.sub(r"\s+", " ", text).strip()
    return text
```

4.4. Skill Extraction Module

The skill extractor checks whether predefined skills are present in the resume text:

```
class SkillExtractor:
    def __init__(self, skills):
        self.skills = sorted(set(skills), key=str.lower)
    def extract(self, text):
        normalized_text = f" {normalize_text(text)} "
        found = []
        for skill in self.skills:
            normalized_skill = normalize_text(skill)
            if f" {normalized_skill} " in normalized_text:
                found.append(skill)
        return sorted(set(found), key=str.lower)
```

4.5. Resume Parsing Module

The resume parser extracts text from PDF, DOCX, and TXT files:

```
def extract_text_from_upload(uploaded_file):
    suffix = uploaded_file.name.lower().split(".")[-1]
    if suffix == "pdf":
        return extract_text(uploaded_file)
    if suffix == "docx":
```

```

        return docx2txt.process(uploaded_file)
    if suffix == ".txt":
        return uploaded_file.read().decode("utf-8",
            errors="ignore")

    raise ValueError("Unsupported file type")

```

4.6. TF-IDF and Cosine Similarity

The resume and job descriptions are converted into TF-IDF vectors and compared using cosine similarity:

```

corpus = [resume_text] + jobs_df["combined_text"].tolist()
vectorizer =
TfidfVectorizer(stop_words="english",
ngram_range=(1, 2)) tfidf_matrix =
vectorizer.fit_transform(corpus)
scores =
cosine_similarity(tfidf_matrix[0:1],
tfidf_matrix[1:]).flatten()
jobs_df["match_score"] = (scores * 100).round(2)

```

4.7. Job Recommendation

Jobs are sorted by match score and the top results are displayed to the user:

```

recommendations = jobs_df.sort_values(
    "match_score",
    ascending=False ).head(top_n)

```

4.8. Skill Gap Analysis

Set difference is used to identify skills required by a job but missing from the resume:

```

resume_skill_set = set(extracted_skills)
required_skill_set = set(required_skills)
matched_skills = sorted(required_skill_set
& resume_skill_set) missing_skills =
sorted(required_skill_set -
resume_skill_set)

```

4.9. Streamlit User Interface

The Streamlit interface allows resume upload, text input, recommendation control, and skill gap display:

```
uploaded_file = st.file_uploader("Upload Resume", type=["pdf",
"docx", "txt"]) sample_resume = st.text_area("Or paste
resume text") top_n = st.slider("Number of
recommendations", 3, 10, 5)
if st.button("Analyze
Resume"):
    extracted_skills = extractor.extract(resume_text)
    recommendations = recommender.recommend(resume_text,
extracted_skills, top_n)
    st.dataframe(recommendations)
```

5. RESULTS AND OUTPUT

The system was tested using sample resume text containing skills such as Python, SQL, Machine Learning, Pandas, and Streamlit.

The application successfully extracted skills, compared them with job descriptions, and generated ranked job recommendations.

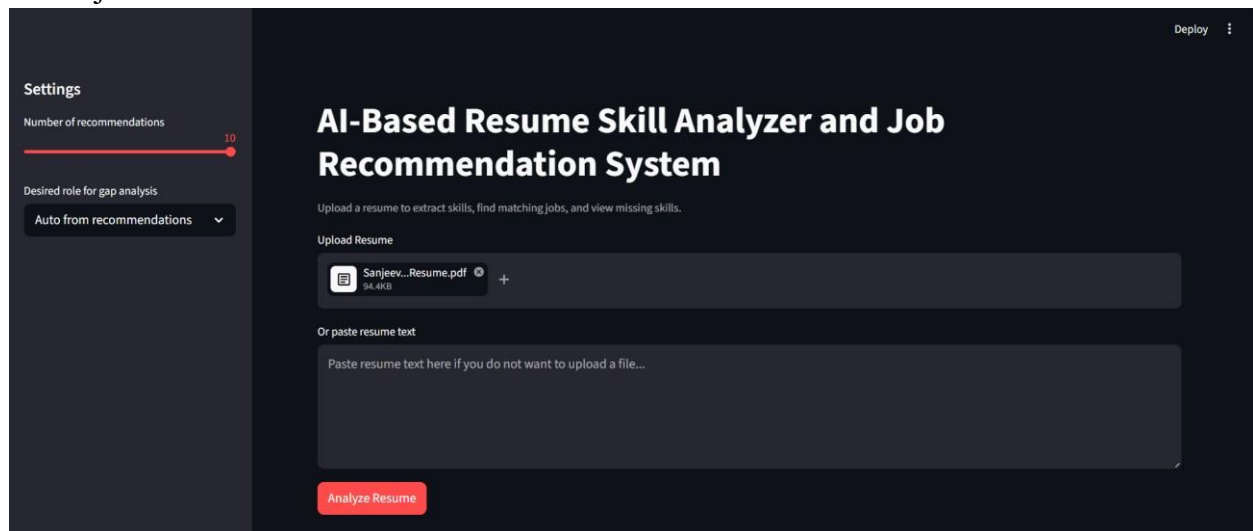


Fig.1 - Output



Fig.2 - Output

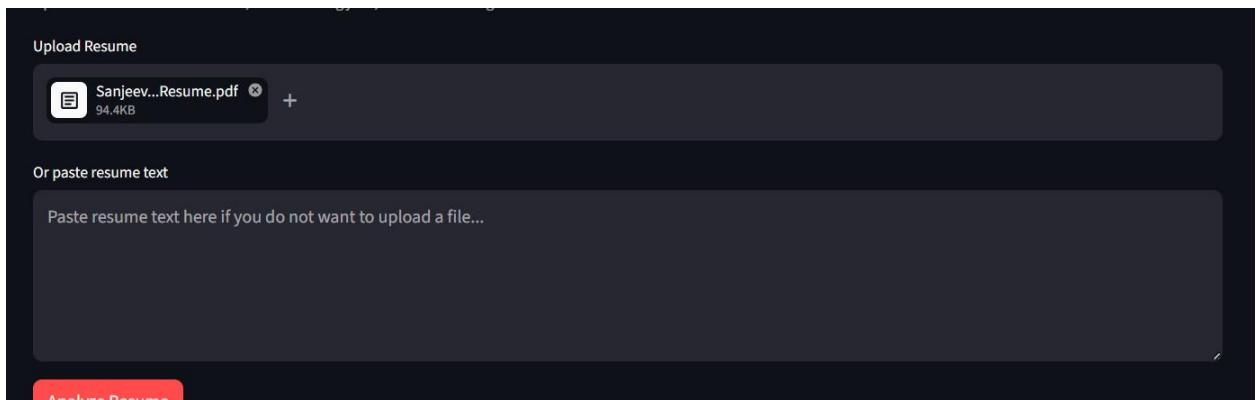


Fig.3 - Output

5.1. Extracted Skills Output

For a sample resume, the system extracted the following skills:

- Python
- SQL
- Machine Learning
- Pandas
- Streamlit

5.2. Recommendation Output

Job Title	Company	Location	Match Score
-----------	---------	----------	-------------

Data Scientist	TechNova Analytics	Bangalore	31.28%
Machine Learning Engineer	AIWorks Labs	Hyderabad	18.35%
NLP Engineer	LanguageGrid AI	Chennai	9.74%

5.3. Skill Gap Analysis Output

For the Data Scientist role, the system compares the user's resume skills with the required skills and displays matched and missing skills.

Matched Skills:

Python, SQL, Machine Learning, Pandas

Missing Skills:

NumPy, Scikit-learn, Statistics, Data Visualization, Model Evaluation, Communication

5.4. Key Findings

- The system correctly identifies technical skills from resume text
- TF-IDF and cosine similarity provide meaningful ranking of job descriptions
- Skill gap analysis clearly shows what the user needs to improve
- The Streamlit interface makes the system easy to use for non-technical users
- The project can be extended using larger job datasets and advanced NLP models

6. CONCLUSION

This project successfully developed an AI-Based Resume Skill Analyzer and Job Recommendation System using NLP and machine learning techniques. The system extracts skills from resumes, compares them with job descriptions, recommends suitable job roles, and identifies missing skills for career improvement.

The key contributions of this project are:

- Automated resume skill extraction
- TF-IDF based text representation for resume and job descriptions
- Cosine similarity based job recommendation
- Set difference based skill gap analysis
- Interactive Streamlit web application for practical use

6.1. Future Work

Integration with live job portals or APIs

- Use of spaCy Named Entity Recognition for advanced skill extraction
- Addition of user login and saved recommendation history
- Use of transformer-based embeddings for semantic job matching
- Personalized learning resource recommendations for missing skills

7. REFERENCES

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